

Active Inference Institute ~ 2024 Quarterly Roundtable #2

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all right hello and welcome it's June 28th 2024 we're here in quarterly Round Table number two for this year welcome to the active inference Institute we're a participatory online Institute that is communicating learning and practicing applied active inference find us at some of the links here this is recorded in an archived live stream please provide feedback so we can improve our work all backgrounds and perspectives are welcome and we'll follow video etiquette for live streams and also I'll show this new slide with some nice stable links that will be good signposts and stable ones too so we're in our second quarterly Round Table of a year there's going to be three main sections First Institute updates then we'll go to the to organizational units and talk about project scale updates and then look a little bit towards the second half of 2024 first though anyone who wants to unmute and just go ahead and introduce say hi and and anything else you want to begin with I'll break the ice hi I'm Blue uh I am really excited at um all of the things that the Institute has accomplished this quarter the fellows program the Partnerships um and our upcoming uh philanthropy program TBD more later I'll go next uh I'm John Bo one of the new fellows this year so excited to be joining The Institute in that capacity and I guess there's more later I'm Jean frer I'm also uh a research fellow part of that first batch and I'll Echo John's words that uh and say I'm very excited uh to uh be in this role and and and hope uh that um I will be able to provide some good value hi uh I'm Andrew um I'm an intern with the Institute I've been involved with a a few different projects on and off over the past maybe year especially helping facilitate the textbook group whenever I can looking forward to that uh coming back up in a couple of weeks and uh yeah looking forward to seeing the rest of the developments with Institute seeing it grow okay good next yeah yeah real quick uh yes so my name is Holly grim and I'm a second uh member of the scientific scientific Advisory board for the second year I'm working with the on the journal project with Alex and U right now I'm kind of building you know a set of Utilities in Python and Russ to help transcribe all of uh Daniel's amazing uh YouTube videos and sessions and uh trying to so

we can eventually create a way of you know using a retrieval augmented generation created an llm agent or uh you know chat with with all of these these doc you know these uh sessions that um have been recorded so thanks okay let me jump in I'm Alex uh one of the officers of the Institute also into Journal project and some hours so great to see everyone let's have a this round table awesome okay okay first going over some Institute scale updates on the digital infrastructure side as the stable links are kind of the tip of the iceberg of we'll be looking and have been making some updates to the accessibility and usability of the different services at The Institute one of those stable links will take you to ecosystem. active influence. Institute and at this page which again the end point may vary will look to develop and really communicate more about what is the active inference Institute what is the ecosystem how are they articulated how how have they begun and changed through time structurally and all these different ways and so one useful piece of information at that resource is kind of like a menu or uh kind of continece of the ecosystem and another exciting announcement is that we'll be doing an updating process on our 2023 Institute ecosystem paper so it'll be very fun there will be a lot of multimedia and Collective exploration and refinements of big impactful visions and things that people are motivated and excited and already working on and so to be able to clarify that within the second half of the year in a certain way and also understand what what is the role of different functionalities in the ecosystem at The Institute so this will be a fun process in the second half of the year that we'll share more information about but it'll involve all aspects of The Institute anyone want to add anything here okay so thanks to the officers the board of directors who had their second quarterly meeting several weeks ago and to the Scientific Advisor board who completed their first phase of activities and we'll head into phase two soon at the website there's more information about these positions for now thanks to everybody who's been carrying it on this quarter one of the really exciting new aspects and programs at The Institute in the last few weeks was the introduction of the research fellows program so perhaps JF and then John what is it in how you're seeing it and what you'll like to do with it I I'll go ahead um well I I've I've been uh conducting a a research project here at the institute for the last uh two years this now called the symbolic co uh cognitive robotics project and this project will now continue uh under the umbrella of research fellowship and uh what I I hope to to gain from this is uh increased uh uh participation involvement exposure which uh uh will bring in unanticipated rewards uh I'm sure it's already started uh and what I hope to provide is uh to use Daniel's words is increased epistemic value uh of my of my research and uh hope to be able to uh present on a more regular basis uh my progress and and hoping that

the Institute will benefit from providing me this this increased exposure um again my name is John Bo and uh my effort is focused on uh cognition in the large group setting so how how can a group of say 100 thousand people or a million people come together in some kind of online ecosystem to make decisions uh uh discuss uh events uh uh and make other cognitive and perform other cognitive processes how does that happen efficiently how can everyone have their Rich input heard uh U and the group in my in my work the group might be just a group of people focused on some issue or it could be actually a something more formal a formal group like a city or U even a nation perhaps one day so that's that's what I'm working on I hope that's will be uh productive and useful awesome and and Anna Pereira will share more about her work in the weeks to come this is an exciting program it's as with all things on a rolling application basis so reach out if you have any questions or to complete the application that's about individuals who are learning and applying active inference also on the interorganizational scale we have begun the part ships program so these are structured relationships between the Institute and other organizations and we expect and prefer this will look like many things already in the one accepted application which will'll announce in the coming weeks and several that are in progress It's really interesting and useful even already to explore where people are at and understand how to respectfully and appropriately have partnership anyone want to add any comments or or thoughts on this all right and one major piece of news from the last several weeks was receiving the 501c3 or USA tax exempt status so the link donate. active inference do Institute is one place that will develop into an on-ramp for providing different kinds of of financial support we're definitely looking to learn and to collaborate and to understand how we can support the sustainability of The Institute and through that support the accessibility and utility of active inference anyone want to make any comments on this I just wanted to thank Fred Frank who was our um they were our pro bono attorney that really worked hard over a long time to uh help us achieve this status thank you blue important to mention too thank you a lot on different digital platforms we continue being active and growing bit by bit as we turn now to the middle of this Round Table where we'll look at project scale updates and through systems engineering and Alex's in particular vision for understanding this hierarchical nested nature of our ecosystem organization and project-based teams in 2024 following Chris fields's information physics as information processing uh course we decided to jump in full F to the prepare measure Paradigm so here's Alice and Bob or system A and B communicating and this could be communication of a thing through time could be across a Communications interface and so we thought about different ways to use

active inference to learn and apply active inference and saw a connection here between the project-based teams and organizational approach and to connect that to the quantum active conference prepare measure cycle and so it's only been half of a year however it's been a very interesting Dynamic already with quite quickly in the beginning of the year and then also continuing people submitted project preparations and and updated whether it was a Institute project that was just kind of reentering the system or projects from people we knew projects from people we didn't know various people completed these project proposal forms and their projects were listed as active and then just in the last several days with many people's measure forms being submitted the rest of this round table and the newsletter which is basically the same information broadly same updates is prepared essentially in a cypaste format from what people have reported at the project scale in their own learning so this is really exciting to see not just the Cod of forms capacity the kind of digital affordances that can be implemented various ways but really the fact that people want to and are submitting more and more parts of their experience on projects of their own other projects and so on so I look forward to this continuing to increase and across multiple scales new and creative uses of the preparation and the measure it's a really just exciting way to kind of not just learn by doing but be doing by doing and that's pretty cool with active inference any general thoughts on this or or anything Quantum active uh yeah let me add something I want to say that from very beginning we we tried to appli First some first principles into organization development and process development starting from active inference systems approach and now even Quantum information processing and it's possibly could look simple like two two new entities prepare measurement forms but I believe as they are based on really like real physics that the current state of the physics it will bring into organization and to Institute a great eff to develop based it thank you Alex very well said anyone else okay we'll just jump into projects so first from the edu active which is education organizational unit just to give a snapshot and separate out the projects into those hosted at The Institute and those happening in the broader ecosystem first in the broader ecosystem Susan Hasty's research on negotiation affordances Ana Magdalena ortado work on UNIF fysica we'll get to these updates Pablo FM and others with num minia and shaa Dobson Hector manrique and Dean and others in the math art conversation so those are projects that facilitators have just stepped up to do and several of the core educational projects at The Institute the active inference ontology audiovisual production textbook group and the journal so let's hear a little bit more about those projects on the textbook group cohort 6 will continue with the second half of the book and cohort 7 will begin from the beginning of

the book on July 15th it'll be a fun go around again it'll have some classic and familiar characters like the ACT INF onology and using different kinds of question-based methods for Learning and asking fun collaborative discussions there'll also be some different elements it'll be on a three-week rhythm we'll do two weeks per chapter and then each of the cohorts will have a third week in the second half of the book that third week will be more about applying active inference looking more at the code which is what the second half of the book is about and in the third week of the first half of the book we'll Explore More what that might look like and also with collaboration with Andrew pendin we will have some additional math learning collaborations and there's a lot of fun continued work on the textbook so it'll be fun to have have had that month or two off jump back into second half of the year with a textbook group any other textbook group thoughts all right in production we've had some fun videos over the last couple months if anyone um wants to call out any specific live stream that they participated in or or viewed and thought was fun go for it there was nine guest streams there was an art stream six insights uh excellently planned and executed by Darius and including the Ian mcchrist surprise which is I think happening concurrently there were several math art streams that were very creative and very exciting we did a paper based live stream series number 57 which was the first in a few months on the paper by par friston and zidan active data selection that was extremely relevant and a really promising direction for the development of active inform and there was an or stream with Nathan Schneider on governable Spaces that was also really cool so anyone want to add any other thoughts okay on other updates from edu active from the fundamentals of active inference project we concluded a phase of work with sanjie in his incredible Odyssey of the fundamentals of active inference we we concluded essentially weekly meetings since the beginning of 2023 and we look forward to next phases for for that project and work Anna Magdalena orta reports from UNIF fysica project and I'm sure we'll hear more from their work Susan Hasty's weekly meetings also came to a excellent closure in the action research on Collective foraging project as she heads into some new programs and directions for the work Pablo FM reported from the num minia project from one of the symposia there was a demonstration of an active inference game in an escape room in the same Spirit has been built which is kind of fun and people can check it out in the newsletter where everything is for all the links and all that from the math art project so kind of in the same Forest as the live streams from math art but part of the broader conversation Bert who continues to do awesome work in the internship made a short report that I know there's so much more to say on and from the journal project Alex and Holly want

to share anything or what what is exciting from Journal yeah so I kind of mentioned it already that I'm interested in uh doing retrieval augmented generation and so um I was able to get um you know a huge uh one tbte um disc from you Daniel and and take all those uh videos and transcribe them uh using a local whisper model on my um on my server here at home so so that was nice in terms of not having to pay too much money out to get those transcripts done by um you know third party uh tooling so and uh yeah and using open Source models so we can start to uh you know make use of those transcripts and I'm pretty excited about that so Alex anything you want to add no great thanks thank you all right to Reiner all right just kind of snapshotting what projects we have that are active so essentially those that have provided us a measurement and any that aren't here can be brought back on by measurement from the ecosystem um Jeremy Cooper and others have been working on active inference account of belief updating in PTSD Michael Carl and others have been working on the active inference model of translations for language JF mentioned the symbolic cognitive Robotics and shager ramen has continued work in writing on the humanity story of Uncertain self at The Institute we continue faster and slower and differently on knowledge engineering and seeing different opportunities for the kinds of knowledge engineering approaches that might benefit active inference in the ecosystem active block fence combining active inference with the cad CAD systems engineering framework by block science is continuing to proceed on a few different directions and then one of our major activities in Reiner has been the RX infer Learning and Development Group so just to sort of contextualize this in active model stream 4.1 implementing active inference by message passing in a factor graph from August [Music] 2021 I looked back to see this presentation by bias lab and see what they had planned to have done for the future of applying reactive message passing which at that point was a research Advanced on top of the factor graph formalism the 40 Factor graph constrained 40 Factor graph and so on had enabled better Logistics to go from the basian graph to the variational message passing and then this reactive feature was getting added in and so it's just really interesting to see how they laid out forny laab 2.0 forny lab. JL 2.0 and then come 2024 we're having great times in the engineering channel in Discord and on a lot of people's personal projects cass's work at learnable Loop people interested in using RX and fur at different kinds of hackathons and different projects and then also a a crossover moment with the textbook group was with the help of some members of the bias laab here's one of the early figures from the 2022 textbook of the frog jumping out of the hands or the something jumping out of the hand or not and then here's what the app model block for RX and fur looks like and so seeing

this kind of an isomorph mapping between the visual representation the code the iCal formalisms the art all these different edges that's what kind of the fun has always been suggested to be so to start to see the operational improvements in reactive message passing and the way that the the groups and the developers involved in this project are closing the gap between what is suggested in principle and in computer science proof of example and taking it to just an amazing engineering informed level anyone want to add anything on aroner I want to say that it's really great development and um it's um good that we organized with group of people who are beginners on learning in that toolbox and I believe that toolbox which uh should provide affordance to build a agents and Collective agents uh will be a base for some breakthrough in artificial intelligence and possibly we are putting some Cornerstone for it thanks Alex and thanks to the the the many participants who are just doing awesome discussion in the engineering Channel contributions to the Koda and to to the GitHub directly all right from the active inference ontology project there have been quite a few updates Dave Douglas as always leading the V of birds has produced two amazing open- Source works that are almost like snapshots of really broad ontological inquiries so first a single author paper aligning spatial web terms to Sumo which is in reference to the itle e p287 4 standards draft and working group connecting that to Sumo which has always been this kind of vision of incorporating multiple ontologies including formal ones like sumo and then also in a snapshot of our primary active inference ontology we had a number of authors join this open- Source snapshot about aligning the active inference ontology to Sumo so starting to triangulate and constellate multiple ontologies along the way learning a lot about using communicating languaging around them and and different functionalities and niches for different structures and visions of onology so that kind of like meta ontological active inference component and also bringing the active inference ontology in a in a connected way to formality while also exploring in the very same project um and and in kind of other meetings and spaces more of the the in formalizable ineffable components so that's pretty cool to see any ontology ideas just a short short note that it's okay to have like M multiple sces multiple scales of uh uh models and have Meta Meta Meta models cool um all right and from other Reiner updates we have Andrew and JF so how about Andrew and then JF if you want to add any more what what have you been uh working on uh yeah and thanks for having me uh so um yeah quick hearkening back to other projects I've been working with Jeremy Cooper on the updating uh belief Dynamics uh it's a computational Psychiatry project Focus ing on PTSD and something I really appreciate about the Institute is its way to kind of remotely bring together uh a variety

of you know people and researchers of different backgrounds that group includes a computational neuroscientist uh a clinical therapist and then you have someone like me and we're we're all learning kind of as we're doing in order to try and recreate something like the belief Dynamics of an individual has PTSD in the direction of developing developing treatments so uh it's it's been a super valuable experience as well as on and off working with the RX and fer group as I continue to develop my own ability to kind of code uh active inference agents and simulations so um so yeah here this project uh active agents and active inference approach to agent-based modeling and the social sciences of be giving that this year at the uh 10th International Conference on computational social science being held at University of Pennsylvania uh basically the tutorial attempts to address a growing interest in more cognitively complex agents to be used in agent-based modeling as opposed to just strict if then uh sort of Agents if this happens then they necessarily do this and so on uh giving them the ability to make inferences and so I've taken a lot of inspiration from um work like uh your own uh work Daniel on the inference project that was my first Fay into like thinking okay I can model an individual agent but what how do you model multiple as they interact an environment that changes which then impacts them getting a better grasp of differentiation between internal and external States um it's been great as well as another uh paper by uh some folks from the scientific Advisory board for the Institute M alberene and Connor Hines um which is also really great very social science oriented uh work um so I'm doing something in turn as kind of they did uh I created a simulation that recreates a common social science paradigm time uh multiple agents working in a network engaging in parallel problem solving and it it using active inference it it recreates the Dynamics of the original Paradigm if you were to run it through net logo or some other common software but uh you can also look under the hood at the agent's own beliefs and I think that this could be used potentially for policy uh development uh it could be used for you know if you have students uh assigned a group project how much do they actually learn to rely upon themselves working in a group as opposed to stealing their neighbors answers um so interesting stuff anyway kind over speaking here but uh thanks for including this in the uh in the talk here so awesome cool and helping facilitate the previous interval of the textbook group too and and with Ali on that too JF sure uh so the uh symbolic cognitive robotics project um asks the question um what does it take at a minimum for a a simple mobile robot to survive in a world it knows initially very little about so the focus is is in on learning through engagement with uh an agent's environment it focuses on the development of agency uh in a robot which starts with very little in terms of uh

cognitive constructs and through engagement through acquisition of experience grows a um a Society of Mind a a collect of simple cognition actors uh which each one uh quite uh specialized in terms of what it can perceive and what it can do um and and they all work together in the sense that um a cognition actor Will through observations of its umelt uh generate synthesize beliefs which will then become observations to more abstract cognition actors so you have this ladder of belief abstraction uh that is built up through active engagement in the environment through uh learning what simple perceptions mean uh learning what affordances there are what can the robot do um and as the robot grows its Collective of cognition actors uh the conjecture is that it will detect um Trends and uh how it's its beliefs its more and more abstract beliefs map uh homeostatic signals am I doing something that benefits my survival in in my continued survival so it's really a project about um uh how a an agent learns its own agency learns its affordances learns to make sense of its perceptions through more and more elaborate beliefs um and and all of this sense making is grounded in the survival imperative now of course the the physical robot is not mortal I mean I can turn it off turn it back on it's still there it's like your computer but the Society of minded will grow through its its experiences through its learning is itself mortal and I set up a an experimental environment such that the robot as such as a minimum amount of time in order to grow competencies at exploring and foraging otherwise it expends its store of energy without ever replenishing it so it needs to forage to replenish it needs also to avoid collisions because collisions are deleterious to its Health it it needs also to keep a level of Engagement otherwise it it it essentially Retreats away from the environment so the all these motivations um uh activate this this this um Society of mine that the robot uh grows over time and all I'm trying to do is set up this mechanisms these mechanisms and launch the robot and see if it can grow it's it's make up its own mind basically about the world it needs to engage uh with so there are Notions of collective intelligence there are Notions of mortal Computing there are Notions of uh grounded sense making uh affordance Discovery um there's also um a very big nod uh to uh Emanuel Kant and it's in his epistemology because the the robot is is needs to develop causal theories of its perceptions how do I make how can I predict what I will perceive next is is very important so a very strong notion of predictive processing based on the discovery of causal Theory how do I make sense uh of of the perceptions and the actions that I that I that I engage into how can it how can I predict what happens next and and all these concepts are are brought together in in in a um in in in a hole that hopefully uh leads to uh autonomous agency and really simple Lego robots which brings

in a dimension of fun as well because these robots are really fun to build and fun to run and my hope is that not only does the robot get to learn how to survive but that we can observe its behaviors and its OB in its visible States and subject them to an active inference analysis and be able to demonstrate that yes indeed the robot reduces its variational free energy and gets better at it over time or not so it's a very fun project and I uh I I I I get a lot of uh uh input from the community and uh I'm constantly fed by all the material that the Institute uh produces so it's um it's a very fun thing to do awesome thank you both for the updates all right that'll be uh sorry yeah go ahead please please yeah as for Andre's work uh I believe at some point it could be integrated into social sciences course somehow if it's interesting possibly it's uh would be not only in our chapter in the course but also somehow argument our chapters with some computational approach and practice and I want to say big thanks for JF for his continuous work on his robotics projects it's really great and really great to see its development through the time thank you very much thank you I agree with that anyone else want to make any comments on Reiner or projects I'd ask a question to JF if I if we have a second can you just talk a little bit about your use the term symbolic in symbolic cognitive yes so the um the causal theories that um each that the the robot discovers and I have to say uh its mind is a society of cognition actors and each cognition actor will want to develop a causal Theory to predict what it will observe next in its own limited umelt those causal theories are essentially logic programs that it discovers uh I will observe C if I've I'm observing A and B for example um so uh much of the learning done by the robot are captured as either um logic programs which are symbolic or and or as uh beliefs which are also uh uh symbolic predicates so uh the the world view that the robot develops internally intrinsically is expressed symbolically as these logic programs and these predicates thus symbolic in what language ah the uh I use two programming languages for the symbolic processing uh and discovery of of of of U causal theories I use prolog which is very well suited uh to uh State uh to to express uh State search problems uh search space uh problems and to solving them and I use a functional Pro language called lexicon to implement uh perception and action basically interfacing with the robots percept uh sensors and uh actuators prolog in Elixir thank you all right as we head into the last part of the Stream So if anyone has questions from live chat they can write it otherwise I will make some notes on one slot and then anyone who wants to bring up anything or ask anything or or anything that they're excited about for the second half the year or however they can go for it so um in the hour before this stream ends and on the verge of a loss of nothing to share I just brought in

some pictures from the prior weeks and wanted to open this space and have the time of reflection and this multiscale self and multiscale time and space Concepts and just where have we been we're in the round table where are we at where are we heading and then where when how Etc is active inference on that and and just a few of The Adventures of the last week well first where where were we in the last end of the last round table and here we had these multiple Horizons and what was happening in active inference and what we could be doing and a a visit to host archival materials and a visit to view Bucky Fuller's archive materials both really impressed upon me the the relevance and necessity of documentation archiving intergenerational culture contact and many more things and this was a funny picture taken from a diner we deliver Norms but we don't only deliver Norms we're also part of them and so on so that was kind of a funny sign to see and then just in terms of where when not traveling where I'm traveling and how to move into the next half of the year so that's all I'll say there how about just anything people want to ask or mention for coming forward I have a question why is your bike called Red Queen oh just kind of moving faster and faster and staying in the same place anyone want to add anything or what yeah Alex yeah I want to say say thanks to all everyone who participated today and that's great that we bring back back to the format of uh participating bro participants then just officers and uh it will be great to see people next rounds so let's continue actually I had a a quick question I I used to be much more involved with the Engineering Group the RX and fur group and un fortunately had to sideline my participation with it have there been any particular ularly interesting developments recently aside from the the live live stream what to say well other than look at the documents and their repos um several things have been leared and developed kass's work at learnable Loop and also the developer teams examples have very rapidly caught up with the major release 3.0 which was a somewhat breaking change in the sense that the prior app models were changed and didn't work in 3.0 but it's built in a better way so to see how fast the developers were able to iterate on the versioning and then the learning of the model building and then to as we start to have people with more computer science less more active inference less and Engineering experience and all this and and all these different perspectives and that helps to disentangle where are the paths that lead into the analytical details of message passing how is that being implemented from what space of ways it could be implemented so like what's required by RX and fur and RX environments what's just a norm of use emerging Norm of use but not something required so a lot of this is still being fleshed out but to see some new types of models open source and then just personally like to be able to take kass' examples copy them

into an online Cloud browser notebook hit run all reload the Page hit run all again and have a real drone active inference simulation schematic but educational run in two clicks it's just something we didn't have one two three Etc years ago so to be and for that to be not just a rapper or like a buffer around a simplified way but that rather to be an interpretable engineering grade spec that's just brilliantly abstracted from the reactive message passing it's a major Tech unlock super cool um also bonus points that uh you can so quickly reproduce the code uh in your own system or in your browser and run it that's what I'm doing for my conference I'm doing everything in Google collab so that attendees can just hop in you have a Google account you can do whatever you want uh I think that's a great sort of Learning Resource for for institute members and others too just keep it keep it ready to go no need to dig through a ton of uh you know library archives and subd directories to finally get the thing running yeah anyone else want to add or ask anything yes since it's my first Presence at this round table I just like to uh Express that um how much I appreciate the openness and and and uh welcoming nature and the welcoming nature of the this community it it really encourages multiple backgrounds multiple perspectives it it listens uh and and it speaks in in very in many voices and I find that it it really uh uh promotes true open science and it's it's I don't take it for granted it is rare and I appreciate it thank you JF okay lasts call okay thank you everyone thanks everybody joining and looking forward to see you around so bye